Guidelines for estimating STM32H5 MCUs lifetime

Introduction

This document describes the estimated lifetime for all the STM32H5 devices. These estimates must neither be taken as actual lifetime limit nor product guarantee on any single device from supplier. Applicable sales and warranty terms and conditions remain in effect.

The data presented in this document is provided for convenience. However, it does not represent all potential failing mechanisms and may not precisely reflect behavior across all mission profiles or applications.

High temperature operating life (HTOL) stress is used for product lifetime assessment trials. Arrhenius equation is used for device aging acceleration factor.

1 General information

The STM32H5 series microcontrollers are Arm®-based devices.

Note: *Arm is a registered trademark of Arm Limited (or its subsidiaries or affiliates) in the US and/or elsewhere.*

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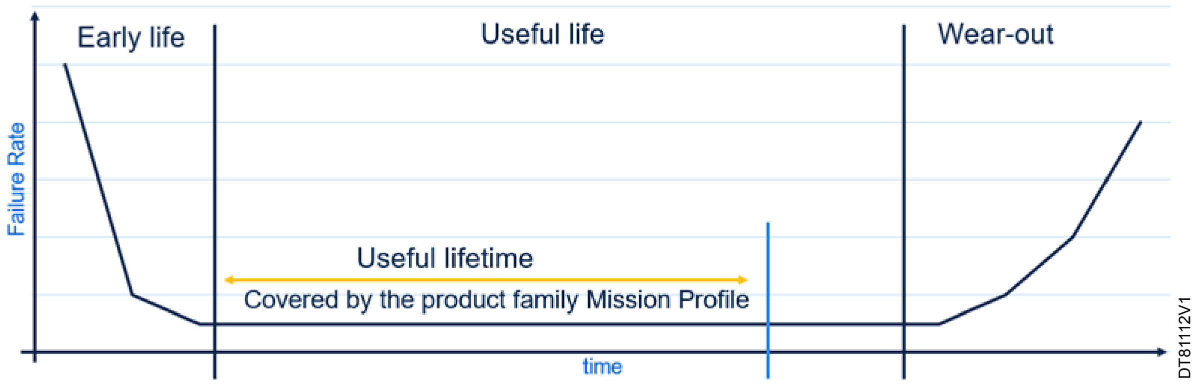
2 Reliability analysis using the bathtub curve model

The bathtub curve model is a widely used reliability concept that describes the failure rate of a device over its lifetime, divided into three distinct phases: Early life, useful life, and wear-out. This model is visually represented by a curve resembling the shape of a bathtub, where the vertical axis represents the failure rate and the horizontal axis represents time.

Figure 1. Bathtub curve concept



- Early life (infant mortality)

Characterized by a higher initial failure rate, which decreases rapidly (expressed in ppm). Usually attributed to manufacturing defects.

- Useful life

Characterized by a relatively constant and low failure rate, which remains stable over the useful lifetime of the device.

- Wear-out

Characterized by increasing failure rate, but normally it should occur much later than the target useful life. The intrinsic wear-out mechanisms begin to dominate, and the failure rate increases exponentially.

- Device useful lifetime

The useful lifetime of a device is the period covered by the product family mission profile, which is demonstrated on silicon.

Table 1. Lifetime phases and failure characteristics

Phase	Failure rate behavior	Causes
Early life	High, then decreasing	Early defects, usually manufacturing
Useful life	Low, constant	Random failures, intrinsic
End of useful life	Increasing	Aging, wear

3 Product mission profile

The product mission profile (MP) defines the various lifetimes available to the device under a given set of conditions. These conditions include the target core voltage (V_{CORE}) and main supply voltage (V_{DD}), the target junction temperature (T_J) of the device, and the operating conditions under which the device operates. For improved reliability and extended lifetime, it is recommended to select a target T_J margin below T_{Jmax}.

- The device operating T_{Jmax} is specified on its product datasheet.
- Operating device at its T_{Jmax} for an extended time reduces device lifetime.
- Effective device thermal regulation must always be ensured that T_J does not exceed what is specified on product datasheet.

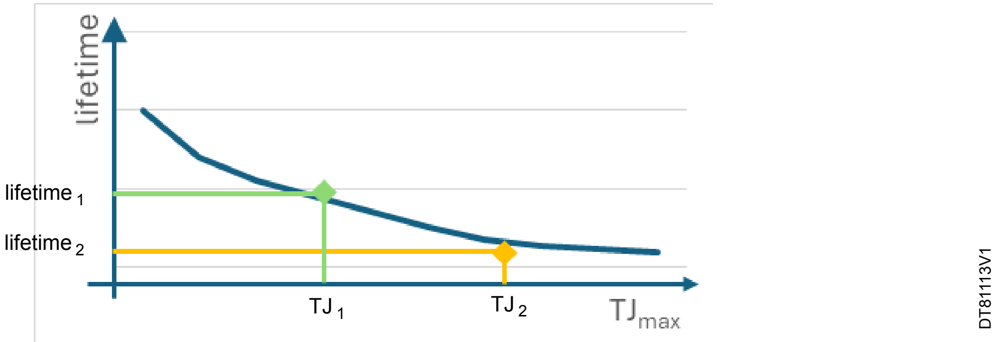
Note: The device useful lifetime data provided in relation to the product mission profiles in this document are intended solely for intrinsic product lifetime estimation.

4 Lifetime curves

The lifetime curve example shown below represents the minimum product lifetime at T_{Jmax} which is assessed by HTOL reliability trial. Any T_J operating point that is lower than the T_{Jmax} operating point is intrinsically covered by this assessment trial. At each T_{Jx} there is a corresponding estimated lifetime.

For example, if $T_{Jmax} > T_{J2} > T_{J1}$, then Lifetime ($T_{Jmax}$) < Lifetime ($T_{J2}$) < lifetime ($T_{J1}$)

Figure 2. Lifetime curve example

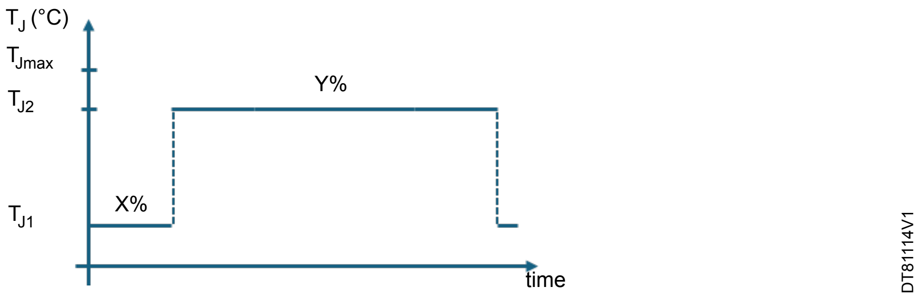


5 Multiple duty cycles with multiple T_J

When T_J of the product does not stay constant during its operation, the target useful lifetime can be calculated using a weighting method based on the duty cycles percentage of the lifetime corresponding to its T_J, which can be obtained directly from the estimated lifetime curve. As products age faster at higher T_J, the combined estimated lifetime is not just a simple average of multiple estimated lifetimes but a value from weighting on duty cycles percentage.

$$\mathrm{Lifetime}_{\mathrm{combined}} = \sum^{n} (\mathrm{Weighting}(\mathrm{Lifetime}_{n}))$$

Figure 3. **Multiple duty cycles and multiple T_J**



The target useful estimated lifetime for a multitemperature use case is calculated with a computation formula with multiple temperatures profile:

Weighting method: Lifetime = (x % at T_{J1}°C) + (y % at T_{J2}°C)

x is the duty cycle at T_{J1}

y is the duty cycle at T_{J2}

x + y =100%

The target useful estimated lifetime = (x % T_{J1} + y % T_{J2})

An example of combined lifetime estimation with multiple duty cycles and multiple T_J values is provided in the next section.

6 STM32H5 series lifetime usage estimation

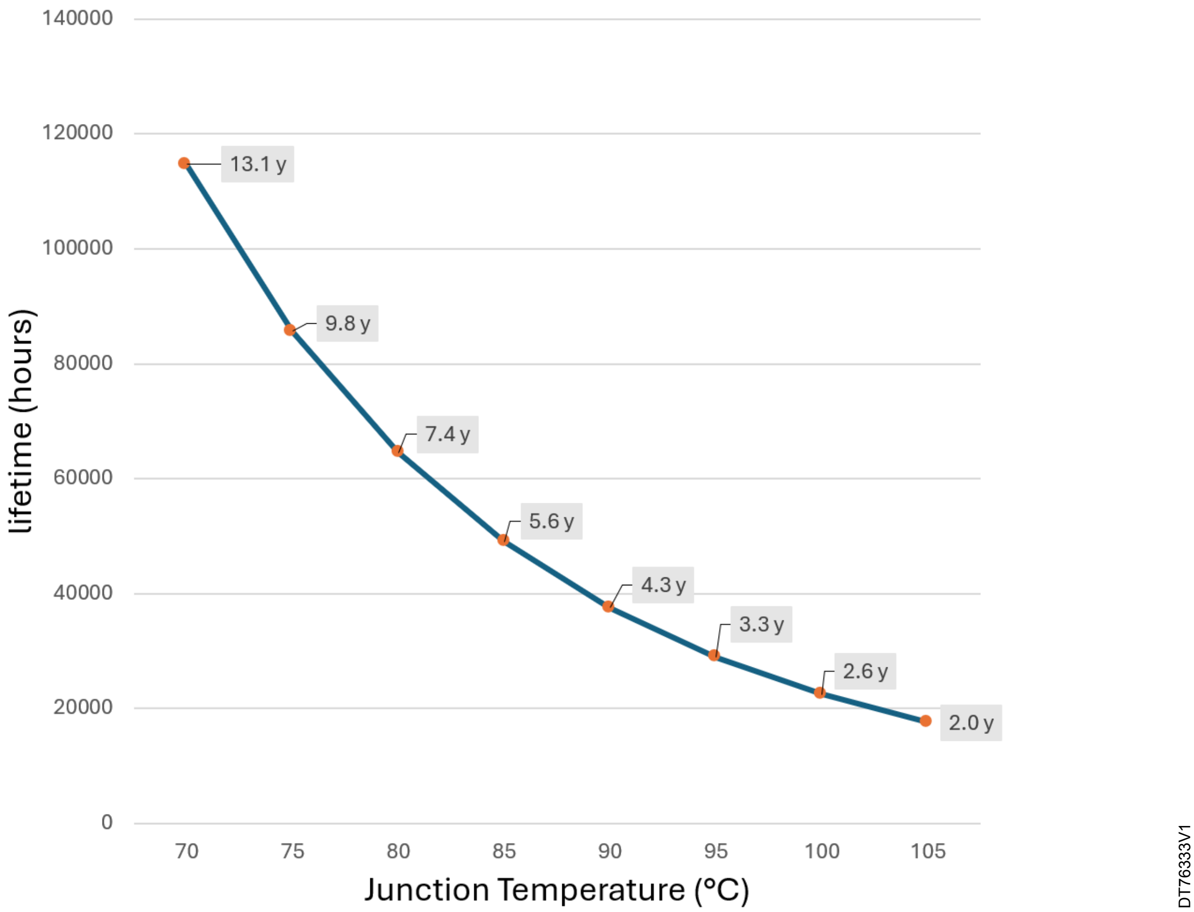
This section provides data and tables representing the lifetime usage estimation for STM32H5 series devices for typical use conditions.

Table 2. **STM32H5 series lifetime usage estimation for typical use condition**

Product MP variant	Lifetime (Years)	Voltage scaling	Duty cycle (%)	V_{DD} nominal (V)	V_{CORE} nominal (V)	Junction temperature (T_J) (°C)
GP0	10	VOS0	20	3.3	1.35	105
IND1	3.2	VOS1	100	3.3	1.2	130
IND2	10	VOS2	100	3.3	1.1	130

6.1 Useful lifetime GP0

Figure 4. **Lifetime estimation GP0**



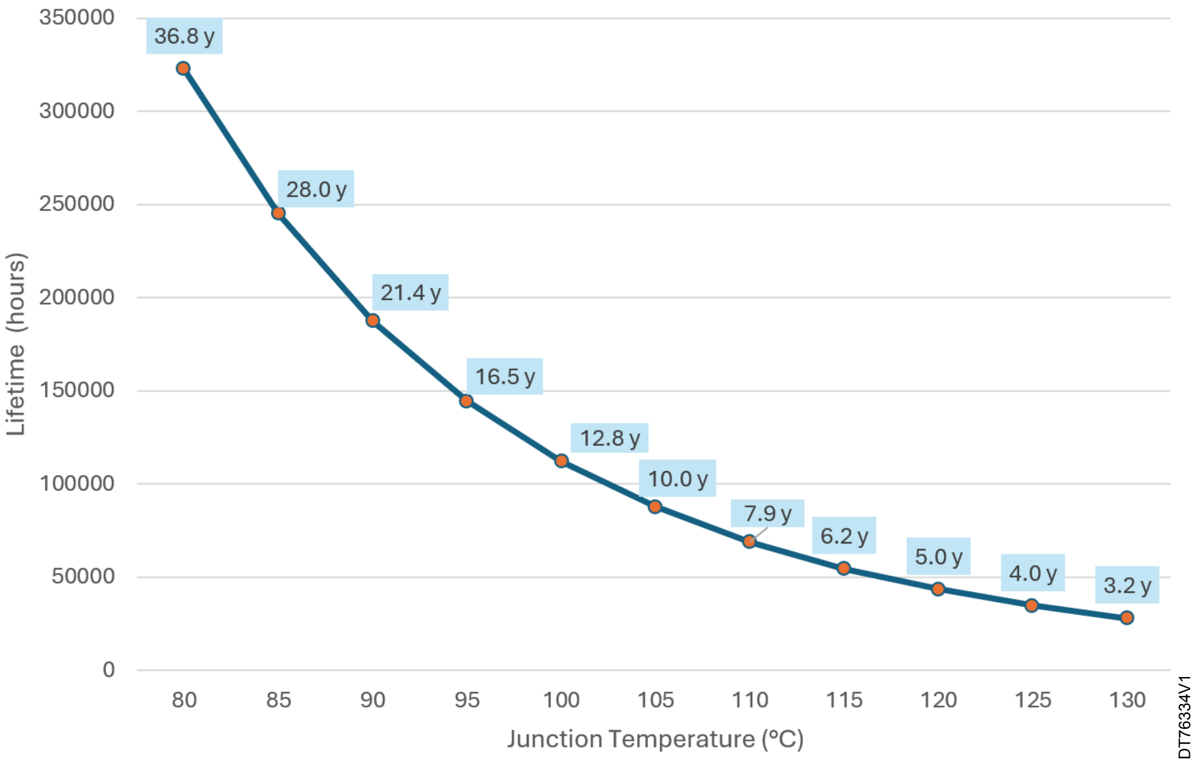
According to Figure 4, when V_{DD} = 3.3 V, V_{CORE} = 1.35 V and a 100% duty cycle, some examples are illustrated below:

- T_j = 105°C the estimated lifetime is 2 years
- T_j = 80°C the estimated lifetime is 7.3 years

Under the same conditions and with duty cycle of 20%, the lifetime estimation is 10 years at T_j = 105°C. This is calculated as (100/20) x 2 years.

6.2 Useful lifetime IND1

Figure 5. Lifetime estimation IND1



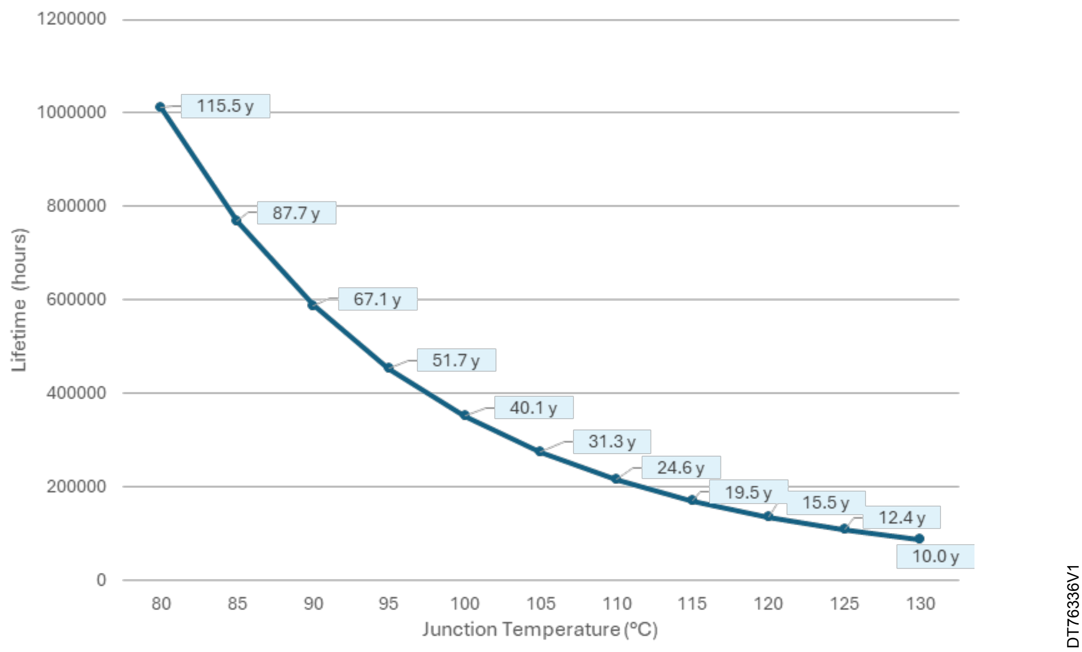
According to Figure 5, when V_{DD} = 3.3 V, V_{CORE} = 1.2 V and a 100% duty cycle, some examples are illustrated below:

- T_j = 130°C the estimated lifetime is 3.2 years
- T_j = 105°C the estimated lifetime is 10 years

Under the same conditions and with duty cycle of 20%, the lifetime estimation is 16 years at T_j = 130°C. This is calculated as (100/20) x 3.2 years.

6.3 Useful lifetime IND2

Figure 6. Lifetime estimation IND2



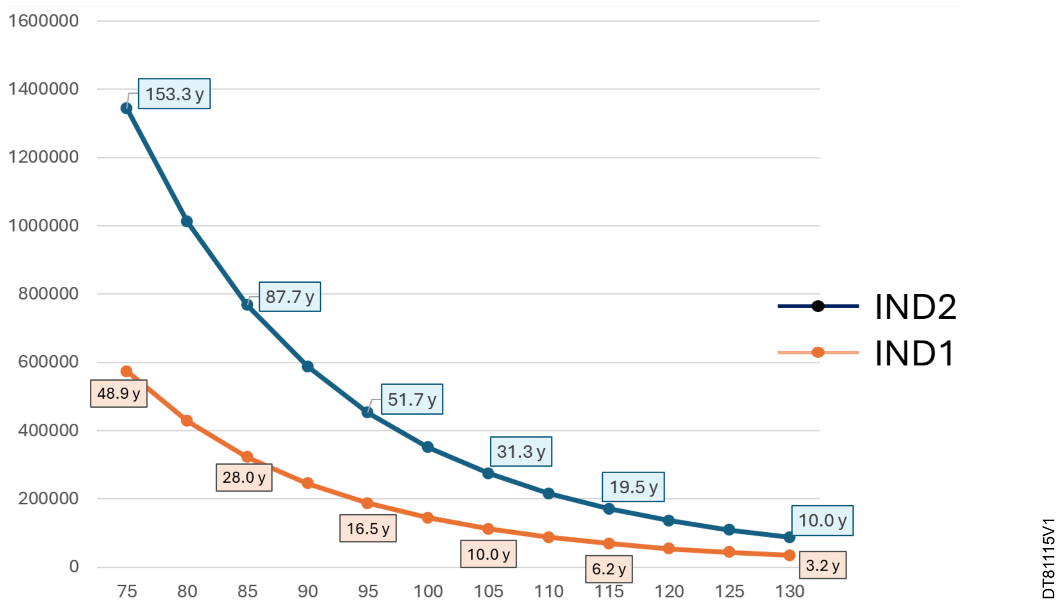
According to Figure 6, when V_{DD} = 3.3 V, V_{CORE} = 1.2 V and operation ratio 100%. Some examples are illustrated such as:

- T_j = 130°C the estimated lifetime is 10 years
- T_j = 105°C the estimated lifetime is 31.43 years

Under the same conditionsand with duty cycle of 20%, the lifetime estimation is 50 years at T_j = 130°C. This is calculated as (100/20) x 10 years.

6.4 Useful lifetime IND1 vs IND2

Figure 7. **Lifetime estimation IND1 vs IND2**



Device lifetime estimates depend strongly on the operating conditions. For example, at 115°, the estimated lifetime is 6.2 years for IND1 (1.2 V) and 19.5 years for IND2 (1.1 V).

6.5 Useful lifetime with multiple duty cycles and T_J

The IND1.1 mission profile is derived from the generic IND1 use case. In this profile, the device operates with a split duty cycle at two junction temperatures instead of a single constant temperature. The overall lifetime is evaluated assuming that the device operates 20% of the time at T_j1 = 105°CT and 80% of the time at T_j2 = 125°C, as summarized in Figure 8. These two operating points are then projected onto the MP1 lifetime curve to calculate the combined lifetime for this mixed-temperature mission profile, as illustrated in Figure 9.

Figure 8. **Example use case of multiple duty cycles and multiple T_J**

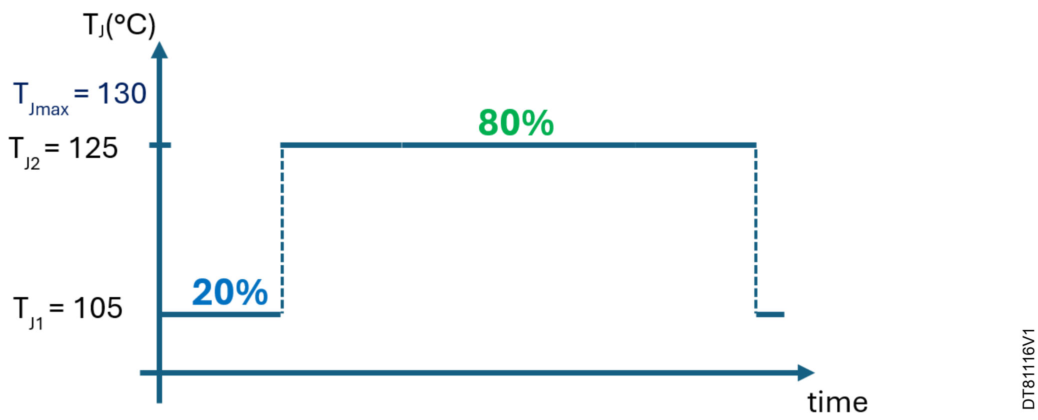
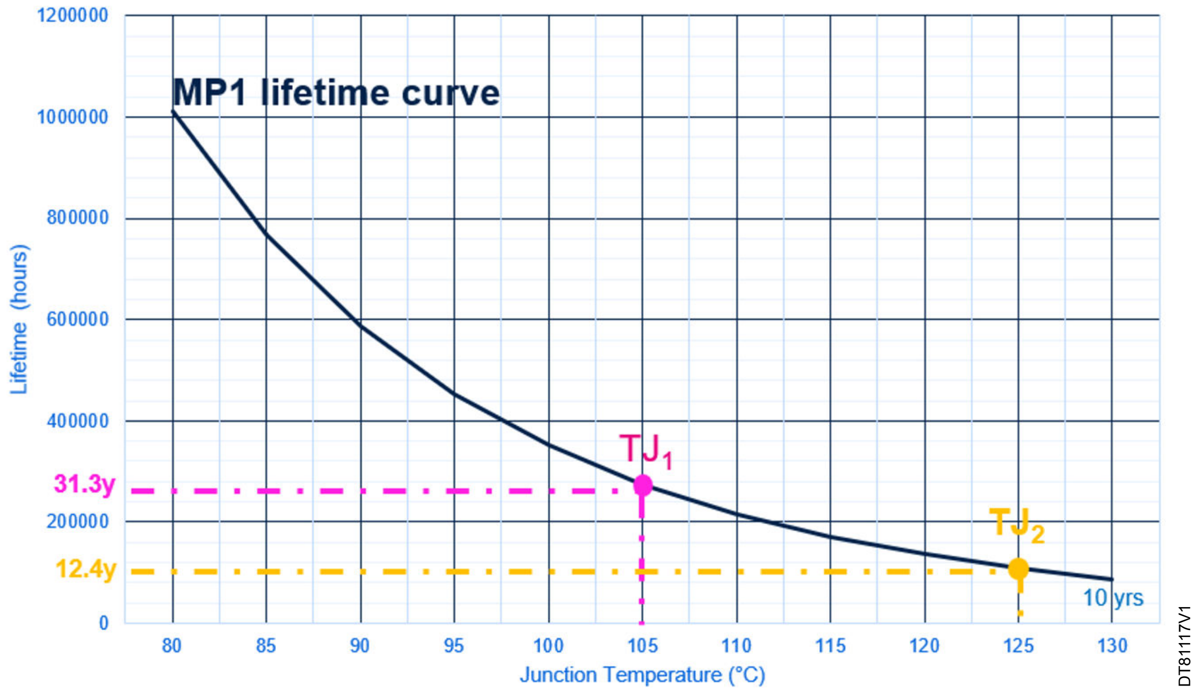


Figure 9. Lifetime estimation IN1.1



As described in Section 5: Multiple duty cycles with multiple T_J, the product is assumed to operate at 20% duty cycle with T_{J1} = 105 °C and at 80% duty cycle with T_{J2} = 130 °C.

Weighting method: Lifetime = (x% at T_{J1}°C + y% at T_{J2}°C)
= (20% at 105°C + 80% at 125°C)

According to Figure 8, at T_{J1} = 105°C, the estimated lifetime is 31.3 years and at T_{J2} = 125°C, the estimated lifetime is 12.4 years.

According to Figure 9, at T_{J1} = 105°C, the duty cycle is 20%, and at T_{J2} = 125°C, the duty cycle is 80%.

The weighted estimate useful lifetime is (0.2 *31.3) + (0.8 *12.4) = 16.1 years

Table 3. Weighted estimate useful lifetime

Product MP variant	Estimate useful lifetime at T_{J1} = 105°C	Estimate useful lifetime at T_{J2} = 125°C	Weighted estimate useful lifetime
MP1.1	31.3 years	12.4 years	(0.2 *31.3) + (0.8 *12.4) = 16.1 years

7 Conclusion

The STM32H5 lifetime analysis shows that, for the specified operating conditions, the required intrinsic useful lifetime can be achieved. Lifetime is mainly driven by junction temperature, and duty cycle: lower T_J, reduced duty cycle, and lower core voltage levels significantly extend lifetime. The curves and formulas provided allow designers to verify that their specific mission profile remains within the demonstrated intrinsic reliability capability of STM32H5 devices.

Revision history

Table 4. Document revision history

Date	Version	Changes
17-Jan-2025	1	Initial release.
03-Jul-2026	2	Updated: • Section Introduction • Section 6: STM32H5 series lifetime usage estimation Added: • Section 2: Reliability analysis using the bathtub curve model • Section 3: Product mission profile • Section 4: Lifetime curves • Section 5: Multiple duty cycles with multiple T_J • Section 6: STM32H5 series lifetime usage estimation • Section 7: Conclusion

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